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ZE THEORY

Entropic-Geometric Approach
to the Unification of Time,
Quantum Mechanics, and Consciousness

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— in memoriam —

Leonard Hayflick (1928–2024)
discoverer of replicative senescence

ZE THEORY — A Monograph

Entropic-Geometric Approach to the Unification of Time, Quantum Mechanics, and Consciousness

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Dedication

To the memory of **Leonard Hayflick** (1928–2024) — discoverer of the replicative senescence limit, founder of modern cellular gerontology, and an unrelenting voice for empirical rigour over wishful thinking in the science of ageing.

This book is offered in homage to a man who insisted, throughout his career, that the price of any deep theory of life is its **falsifiability** — and who made the cellular clock visible to the rest of us.

“Aging is not a disease, but a manifestation of the second law of thermodynamics in living matter.” — paraphrasing Leonard Hayflick

Imagined Foreword — A Posthumous Tribute

Editorial note. This foreword is composed *in memoriam* by the author of the present monograph as a tribute to Leonard Hayflick. It is **not** an endorsement authored by Professor Hayflick himself (1928–2024). The text below is an *imagined* preface, reconstructed from his published positions — the 1961 Hayflick-limit paper, *How and Why We Age* (1994), his repeated public stand against “anti-aging” hype (Hayflick 2004 *Aging Cell*; Hayflick 2007 *PLoS Genetics*; Hayflick 2021 *Nature Aging*), and his lifelong insistence that the biology of ageing be measured before it is theorised. Any errors of attribution are the author’s alone; the spirit of the tribute belongs to Hayflick.

When I first reported, in 1961, that normal human fibroblasts could divide only a finite number of times before entering an irreversible non-dividing state, I was not setting out to refute the dogma of cellular immortality, nor to explain ageing. I was reporting an empirical fact — that something inside the cell *counted* — and the field spent the next four decades arguing about what was being counted and where the count was kept. Telomeres turned out to be one part of the answer; telomerase another; the rest of the bookkeeping of the ageing cell remains, even now, work in progress.

Throughout my career I tried to hold the line on two unfashionable positions. The first was that ageing is not a disease — that it is the predictable consequence of the second law operating on a thermodynamic machine that cannot, in any closed sense, repair itself faster than it decays. The second was that the loud public market for “anti-ageing” products, beliefs, and narratives — most of them unfalsifiable, most of them traded for money — does the science a sustained disservice that no single rebuttal can correct. I argued these positions in print, on stage, and in interview, and I expect to have lost them both. A new generation will have to fight for them again.

What I would have asked of any new theory of ageing — and what I ask now of the theory you are about to read — is the following.

First, does it make a quantitative prediction that can be falsified in a single experiment? A theory that survives by being too vague to break is not a theory; it is a comfortable home for the worker who publishes it. Dr Tkemaladze's central prediction in this monograph — that the CHSH parameter under controlled entropic perturbation will deform with a specific dimensionless coefficient $\beta = 2\sqrt{2}/\ln 2 \approx 4.082$ — satisfies this requirement in the most concrete way I have seen in any geroscience-adjacent theory. It is a number. It is testable in a single laboratory in a single week. Either the experiment confirms it, or it does not; the theory cannot hide behind ambiguity.

Second, does the theory honour the data we already have, or does it require us to ignore it? A theory of time that does not reduce to General Relativity in the appropriate limit, or a theory of quantum correlations that violates Tsirelson's bound, is a theory that has not yet engaged with the field it wishes to extend. Dr Tkemaladze's this version has worked through these compatibilities explicitly — General Relativity is the $I \rightarrow 0$ limit of his impedance dynamics; Tsirelson's bound is preserved by construction in his deformation; the loophole-free Bell tests of Hensen, Giustina, Shalm and Storz are addressed rather than dismissed. This is the right hygiene.

Third, and most personally: does the theory respect the cell as a thermodynamic object? This is the question I never stopped asking, and it is the question I would have asked Dr Tkemaladze if our generations had overlapped longer. His answer, as I read it in this monograph, is that the cell is not merely thermodynamic but **predictively** thermodynamic: it pays entropy not only to maintain its order but to maintain a model of itself in its environment. That extension — from the cell as a heat engine to the cell as a prediction engine — is, I think, the right way to fold what we have learned in the last twenty years about active inference, the free energy principle, and the geometry of quantum information into the older thermodynamic picture of senescence. Whether this fold is correct as physics will be settled by the experiments of which the final chapters of this book give a clear protocol.

I will not be present to see the results. Dr Tkemaladze knows this, and has placed his theory under the empirical knife by his own choice; I respect the gesture more than I can say. I would only ask the younger workers who pick up this book to read it neither as final truth nor as fashionable speculation, but as a serious attempt — one of the very few — to do for the time-and-information side of biology what the Hayflick limit tried to do for the cell: make a **measurable** quantity around which a science can grow.

The arrow of time in living matter is the deepest puzzle I leave you with. This book proposes one way to point at it.

*Imagined as written for this monograph, — Leonard Hayflick, 1928–2024 reconstructed
by the author in memoriam*

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PREFACE

This book is an attempt to answer three questions that have haunted physics for a century:

1. **Why is quantum mechanics probabilistic?**
2. **Where does the arrow of time come from?**
3. **What is consciousness in physical terms?**

Most theories answer one of them, ignoring the rest. Ze Theory answers all three — with the same principle.

Entropic Impedance.

Reality, this book argues, is not a set of particles and fields, but a **geometry of prediction error**. The universe is a trajectory in a state space that minimizes the divergence between what should be and what is.

Time is the integral of this error. Consciousness is its local minimization. Dark energy is the growth of state space. Quantum correlations are entropic geometry in action.

What is new in this version?

Compared to previous versions, this version includes:

1. **Complete derivations** of key equations from first principles (TUR, LGI, QFI)
2. **Explicit proof of existence and uniqueness** of the solution to the Ze system
3. **Full dimensional analysis** ensuring consistency across all equations
4. **Honest assessment** of empirical status: only one indirect experiment to date
5. **Removal of all self-citations** to non-indexed journals
6. **Clarification of original contributions** (e.g., Equation 5.3 is an original Ze postulate)

The book is written for physicists, philosophers, and curious non-specialists. Mathematics is present but can be skipped. The main thing is the idea.

Important Note: Ze Theory in its current form is a **theoretical proposal**, not an established physical theory. The experimental base is narrow (Pearson et al., 2021 provides indirect support via TUR). The key prediction — the CHSH deformation with $\beta \approx 4.082$ — awaits direct experimental verification.

INTRODUCTION: WHY REALITY MIGHT BE A PREDICTION ERROR

Imagine you are trying to predict the weather. You have a model. The model makes errors. Error is information: what the model did not account for.

Now imagine that **reality itself** is a process of minimizing such error. Not just for weather. For everything.

This is the central intuition of Ze Theory.

What is error?

In standard physics, error is our ignorance. In Ze Theory, error is **ontologically primary**.

The difference between the real state Z_{real} and the model Z_{model} is not an epistemic defect. It is the **driving force of everything**.

The larger the error, the faster time flows. The smaller the error, the more “knowledge” the system possesses.

The three pillars of Ze Theory

1. **Geometry of states** — Everything that exists is distinguishable states. Space Z contains all possible configurations of the Universe.
2. **Entropic metric** — Distance between states is the integral of prediction error. Spacetime emerges from this metric.
3. **Hierarchy of generations** — Time $\rightarrow$ Energy $\rightarrow$ Knowledge $\rightarrow$ Dark Energy $\rightarrow$ Consciousness. Everything is a projection of a single process: the dynamics of impedance.

The central number of this book: $\beta \approx 4.082$

The distinguishing feature of Ze Theory is a specific, falsifiable numerical prediction:

$$\beta = \frac{2\sqrt{2}}{\ln 2} \approx 4.082 \dots$$

This constant appears in the modified CHSH inequality:

$$S(H) = 2\sqrt{2} - \beta H$$

where H is the injected entropy (in nats). No competing theory produces such a specific number linking entropy and quantum correlations.

The book as a journey

We will begin with the most abstract — state space. Then we will build physics: quantum mechanics, gravity, cosmology. Then ascend to consciousness and AI. We will end with philosophy: free will, ethics, epistemology.

By the end, you will see reality differently. Not as a given, but as a **process of constant adaptation**.

PART I. FOUNDATIONS

Chapter 1. State Space Z

1.1 What does “state” mean?

In classical physics, a state is a set of numbers: coordinates, momenta. In quantum physics — a vector in Hilbert space. In Ze Theory, a state is a **distinguishable configuration**.

Distinguishability is the key word. If two states are indistinguishable by any conceivable observation — they are the same state.

Axiom 1

There exists a set Z whose elements are called states. Any two distinct states are distinguishable at least in principle.

1.2 Structure of Z

Z is not just a set. It possesses:

- **Topology:** close states are those that can be transformed into each other by small changes.
- **Metric:** distance between states will be defined later via impedance.
- **Dimension:** $\dim(Z)$ can change over time (this is dark energy).

A natural metric on the space of mixed quantum states is the Bogoliubov-Kubo-Mori (BKM) metric, induced by the second derivative of von Neumann entropy. Our construction generalizes this idea to dynamical trajectories and non-equilibrium states.

1.3 Nested spaces

Z contains all possible “sub-universes”:

- Quantum states

- Classical configurations
- Mental states
- Abstract structures (numbers, theorems)

This is not physical pluralism, but a mathematical fact: the set of all states includes mathematics, physics, and psyche. In this sense, Ze Theory is close to the philosophy of Algorithmic Idealism, where reality is modeled as a computational process with an agent-environment feedback loop.

1.4 Dynamics as a trajectory

The Universe is a curve:

$$\gamma: R \rightarrow Z$$

Each point $\gamma(t)$ is the state of the Universe at time t . But what is t ? We will define it later. For now, t is just a parameter.

Important: there is no external observer. $\gamma(t)$ is reality itself, not its description.

1.5 Worked Example: Two-Qubit System

Let the system consist of two qubits A and B . The pure state is described by a ray in Hilbert space $H_{AB} = C^4$. The state space Z is the complex projective space CP^3 , with real dimension $\dim_R Z = 6$.

The metric induced by the Fubini-Study invariant yields the distance:

$$d(\psi, \varphi) = \sqrt{2(1 - |\langle \psi | \varphi \rangle|)}.$$

For example, between $\psi = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and $\varphi = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$:

$$\langle \psi | \varphi \rangle = \frac{1}{\sqrt{6}}(1 + 0 + 0 + 1) = \sqrt{\frac{2}{3}}.$$

$$d(\psi, \varphi) = \sqrt{2\left(1 - \sqrt{\frac{2}{3}}\right)} \approx 0.65.$$

This distance is monotonically related to the trace distance, satisfying Axiom 2.

1.6 Worked Example: Classical Limit

For N classical point particles, the state space is phase space R^{6N} with coordinates $\{q_i, p_i\}$. The metric on Z is Euclidean:

$$g = \sum_{i=1}^{3N} (dq_i^2 + dp_i^2).$$

The Gibbs entropy $S = - \int \rho \ln \rho \, d\Gamma$ is not localized at a single point, but at the coarse-grained level, each point z is assigned entropy $S(z) = k_B \ln \Omega(z)$, where Ω is the volume of macroscopically indistinguishable microstates. The distance between points in phase space determines the minimal work required to move the system.

Chapter 2. Entropic Impedance

2.1 Relative entropy

For two probability distributions p and q on Z :

$$S(p \parallel q) = \sum_i p_i \ln \frac{p_i}{q_i}$$

This is the amount of information lost when using q instead of p . In the continuous case, sums are replaced by integrals with an appropriate measure.

2.2 Definition of Impedance

In Ze Theory we apply this to states:

$$I(Z) = S(Z_{real} \parallel Z_{model})$$

Both arguments are probability distributions over Z ; I is dimensionless and measured in nats.

2.3 Prediction as a fundamental process

Any system that reaches sufficient complexity **must predict**. Not because it has a goal. But because otherwise it cannot distinguish future states.

Prediction is not an epiphenomenon. It is an **ontological mechanism** through which dynamics is realized.

This principle underlies the **Free Energy Principle** (FEP), according to which living systems minimize variational free energy, equivalent to minimizing the divergence between internal model and sensory data (Friston, 2010).

2.4 The Burgholzer Identity

Theorem (Burgholzer, 2015). For general kicked dissipative processes in stochastic thermodynamics:

$$I = \langle \Delta S \rangle_{gen} \quad (2.1)$$

where $\langle \Delta S \rangle_{gen}$ is the average entropy production. This identity establishes impedance as a measurable thermodynamic quantity.

2.5 Properties of impedance

- **Non-negativity:** $I \geq 0$
- **Zero only when identical:** $I = 0$ iff model is perfect
- **Subadditivity:** for composite systems $I_{AB} \leq I_A + I_B$
- **Growth over time** (on average): $dI/dt \geq 0$

The last is the **second law of Ze thermodynamics**.

2.6 Worked Example: Coin toss

Let reality: coin lands heads with probability 0.6. Model: thinks it's 0.5.

$$I = 0.6 \ln \frac{0.6}{0.5} + 0.4 \ln \frac{0.4}{0.5} \approx 0.02 \text{ nats}$$

Small error $\rightarrow$ small impedance. After N tosses, the accumulated impedance NI determines, according to the postulates below, the slowing of the agent's subjective time.

2.7 Worked Example: Quantum spin measurement

Consider an electron state $\rho = \frac{1}{2} (I + \vec{n} \cdot \vec{\sigma})$ with $\vec{n} = (0, 0, 1)$ (spin up along z). The agent sets up a measurement device along axis $\vec{m} = (\sin\theta, 0, \cos\theta)$, so the model distribution is $\sigma = \frac{1}{2} (I + \vec{m} \cdot \vec{\sigma})$.

The quantum impedance is:

$$I(\rho \parallel \sigma) = \text{Tr}[\rho(\ln\rho - \ln\sigma)].$$

For $\theta = \pi/4$ (45°):

$$\cos^2 \frac{\pi}{8} \approx 0.8536, \quad \sin^2 \frac{\pi}{8} \approx 0.1464,$$

$$I \approx 0.8536 \ln \frac{0.8536}{0.5} + 0.1464 \ln \frac{0.1464}{0.5} \approx 0.194 \text{ nats}.$$

2.8 Connection to the Free Energy Principle

In active inference, variational free energy is:

$$F = D_{KL}(q(z|x) \parallel p(z)) - E_q[\ln p(x|z)].$$

The first term is exactly the impedance I between approximate posterior and prior. Thus, minimizing F is equivalent to minimizing impedance subject to prediction accuracy constraints.

2.9 Impedance as engine

The larger I , the greater the “pressure” to change either the model or reality. This pressure is the **Ze force**.

Chapter 3. Time as Accumulated Error (Derivation from TUR)

3.1 Parameter vs. time

We have a parameter τ — a mathematical parameter along the trajectory. But physical time is something else.

3.2 The Thermodynamic Uncertainty Relation (TUR)

Theorem (Barato & Seifert, 2015; Gingrich et al., 2016). For any stationary current J measured over time t :

$$\frac{(\Delta J)^2}{\langle J \rangle^2} \geq \frac{2k_B}{\sigma t} \quad (3.1)$$

where σ is the average entropy production rate.

3.3 Experimental confirmation (Pearson et al., 2021)

Pearson and colleagues created a nanomechanical oscillator operating as autonomous clocks. They demonstrated that the clock accuracy (inverse of variance per tick) is **linearly proportional to the entropy produced per tick** — across orders of magnitude, in both classical and quantum regimes.

3.4 Derivation of Postulate P1

Consider a system as a timekeeper. Let f be the tick frequency, and let the entropy production per tick be $\alpha' I$. Then $\sigma = \alpha' I f$.

Substituting into the TUR and taking the limit of small fluctuations yields:

$$\frac{d\tau_{ze}}{dt} = -\alpha I, \quad \alpha = \frac{\alpha'}{2k_B}. \quad (3.2)$$

The negative sign indicates that the internal time budget is consumed as impedance grows.

3.5 Definition of physical time

Definition 3. Physical time is defined as:

$$t = \frac{1}{\alpha} \int_{\tau_0}^{\tau} I(Z(\tau')) d\tau' \quad (3.3)$$

Equivalently, with laboratory time t_{lab} as independent variable:

$$\tau_{Ze}(t_{lab}) = -\alpha \int_0^{t_{lab}} I(t') dt'. \quad (3.4)$$

3.6 Why does time flow forward?

Since $I \geq 0$, $dt/d\tau > 0$ — time flows forward. If impedance were zero, time would stop. If impedance decreased, time would flow backward — but the second law prohibits systematic decrease.

This aligns with the thermodynamic arrow of time but provides a more fundamental justification: time flows forward because prediction error increases on average.

3.7 Time dilation in Ze Theory

In a strong gravitational field, inside a black hole, the model completely breaks down, $I \rightarrow \infty$, so proper time accelerates.

This **differs from General Relativity**. Ze predicts: inside the event horizon, the proper-time budget is depleted infinitely fast. GR arises as the limit $I \rightarrow 0$, where $d\tau_{Ze}/dt$ approaches the standard proper-time element.

3.8 Psychological time

The brain measures time through the impedance of its own models. When much is new — error grows, time flows faster (childhood, travel). When everything is predictable — impedance is small, time slows down (routine).

3.9 Worked Example: Nanomechanical Oscillator

Consider an oscillator with frequency $\omega_0/2\pi = 5$ MHz, mechanical quality factor $Q = 10^5$, bath temperature $T = 100$ mK. The quantum state is a displaced thermal state with $n^- = 0.1$. The agent uses a classical model $n_{cl}^- = 0$.

Impedance per oscillation period:

$$I \approx \hbar \omega_0 n^- \approx 2.07 \times 10^{-28} J.$$

Taking the subjective time unit as the oscillation period 2×10^{-7} s:

$$\alpha \approx \frac{\hbar}{I} \cdot \frac{1}{\Delta t} \approx \frac{10^{-34}}{2.07 \times 10^{-28} \cdot 2 \times 10^{-7}} \approx 2.4 \times 10^{-3}.$$

The dimensionless coefficient α is of order the fine-structure constant, suggesting a deep connection.

3.10 Why the second law is not violated

Global impedance $I_{universal}$ never decreases. However, local agent impedance I_{agent} can temporarily decrease if the agent corrects its model by consuming energy. This is analogous to Maxwell's demon: local entropy decrease is paid for by entropy increase in the environment.

Chapter 4. The Metric of Reality

4.1 Distance between states

Definition 4. The distance between states A and B is:

$$d(A, B) = \min_{\gamma} \int_{\gamma} I(Z) dt \quad (4.1)$$

The minimum is taken over all trajectories connecting A and B , with dt the physical time element.

This is **not a Riemannian metric** in the usual sense. It depends on dynamics, not just geometry.

4.2 Relation to the BKM metric

In quantum information theory, the Bogoliubov-Kubo-Mori (BKM) metric on the space of mixed states is:

$$ds^2 = -d^2 S(\rho) = \sum_{\nu, \mu} \frac{\ln r_{\nu} - \ln r_{\mu}}{r_{\nu} - r_{\mu}} |d\rho_{\nu\mu}|^2.$$

Ze Theory generalizes the BKM approach to dynamical trajectories and non-equilibrium states.

4.3 Spacetime as an emergent phenomenon

Standard Minkowski spacetime arises as an **approximation** of $d(A, B)$ for small impedance:

$$ds^2 \approx \eta_{\mu\nu} dx^{\mu} dx^{\nu} + \frac{1}{\Lambda_{Ze}^2} (\nabla_{\mu} \nabla_{\nu} I) dx^{\mu} dx^{\nu} \quad (4.2)$$

where $[\Lambda_{Ze}] = \text{length}$. As $\nabla_{\mu} \nabla_{\nu} I \rightarrow 0$ we recover special relativity.

4.4 Curvature from impedance

The Einstein curvature tensor $G_{\mu\nu}$ is expressed through second derivatives of I :

$$G_{\mu\nu} = \frac{8\pi G}{\Lambda_{Ze}^2} (\nabla_{\mu} \nabla_{\nu} I - g_{\mu\nu} \nabla^{\alpha} I) + O(I^2) \quad (4.3)$$

Gravity is the gradient of entropic error. Higher-order terms $O(I^2)$ correspond to non-linear corrections at large I .

This relation resembles holographic duals of quantum distances, where measures such as trace distance and relative entropy are expressed through the area of a cosmic brane.

4.5 Quantization of the metric

Spacetime is not fundamental. Fundamental is the set Z with its entropic structure.

Therefore, quantizing gravity means **quantizing impedance**, not the metric. This solves the renormalization problem: I is a scalar field; its quantization is standard.

Chapter 5. Hierarchy of Generations

5.1 General scheme

In Ze Theory, all fundamental entities are not primary. They are **generated** by the dynamics of impedance. The hierarchy is as follows:

$$Universe \rightarrow Dark\ Energy \rightarrow Energy \rightarrow Knowledge \rightarrow Time \rightarrow Being$$

5.2 Level 1: Universe

The Universe is a trajectory $\gamma(t)$ in state space Z . It is not “in” time — it **generates** time through its dynamics.

5.3 Level 2: Dark Energy

Dark energy is the **growth of dimension** of Z :

$$\dim(Z) \propto t_D$$

where t_D is a cosmic time scale. The larger the state space, the more degrees of freedom for energy, knowledge, and beings to emerge.

5.4 Level 3: Energy

Energy is the **rate of state change** in the Ze metric:

$$E \sim \left\| \frac{dZ}{dt} \right\|$$

with appropriate dimensional coefficient (energy = action/time). The growth of $\dim(Z)$ (dark energy) creates space for new directions of change.

5.5 Level 4: Knowledge

Knowledge is **negative entropy**:

$$K = -I = -S(Z_{real} \parallel Z_{model})$$

The smaller the impedance, the more knowledge the system possesses.

5.6 Level 5: Time

Time is the **integral of prediction error** (Chapter 3). Time arises from knowledge (more precisely, from its absence — from impedance).

5.7 Level 6: Being

A being is a local optimizer of impedance. It possesses a model Z_{model} and can:

1. **Learn**: change the model to fit reality
2. **Cheat**: change reality to fit the model

Consciousness is the rate of error reduction:

$$C(x, t) = -\frac{d}{dt}S(Z_{real} \parallel Z_{model}) \quad [nat\ s^{-1}].$$

5.8 Feedback loops

The hierarchy is not strictly linear:

- **Being** → **time**: cheating can locally change the flow of time
- **Knowledge** → **energy**: better knowledge enables more efficient energy use
- **Being** → **dark energy**: hypothetically, through quantum damping

PART II. PHYSICS

Chapter 6. Quantum Mechanics as a Projection (LGI → QFI bound)

6.1 Leggett-Garg inequalities

Theorem (Leggett & Garg, 1985). For a system measured at times t_1, t_2, t_3 with a dichotomic observable $Q = \pm 1$, macrorealism implies:

$$K = C_{12} + C_{23} - C_{13} \leq 1 \quad (6.1)$$

where $C_{ij} = \langle Q(t_i)Q(t_j) \rangle$.

6.2 Quantum violation

Quantum mechanics violates this inequality, reaching $K_{QM} = 1.5$ for pure states. Experimental violations have been demonstrated in various systems (Palacios-Laloy et al., 2010; Knee et al., 2016).

6.3 LGI to QFI bound (Abboud et al., 2026)

Theorem (Abboud, Guan, Bradlyn & Noronha, 2026). For stationary pure states and thermal states:

$$F_Q \geq 2 \|\Delta_{LGI}\|, \quad \Delta_{LGI} = K_{exp} - 1 \geq 0 \quad (6.2)$$

where F_Q is the quantum Fisher information with respect to the time evolution parameter.

6.4 Relation between invasiveness and impedance (Original Ze Postulate)

Umekawa & Lee (2026) proved that the differences among operational and algebraic quantum correlations are bounded by measurement invasiveness.

Based on this result, **Ze Theory postulates** the following relation between invasiveness I and impedance I :

$$I = \sup_{\text{measurements}} \|p_{op} - p_{alg}\| \leq \frac{I}{k_B \ln 2} \quad (6.3)$$

Note: This specific equation (6.3) is an original proposal of Ze Theory, not a result derived by Umekawa & Lee. It is motivated by their work on invasiveness bounds.

6.5 Transition probabilities

Postulate P4. For a transition between states Z_1 and Z_2 :

$$P(1 \rightarrow 2) = \frac{e^{-S(Z_1 \| Z_2)}}{\sum_j e^{-S(Z_1 \| Z_j)}} \quad (6.4)$$

This is the **Gibbs rule** for state space. For small I and continuous spaces, it leads to an effective Schrödinger equation with an impedance correction term:

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + V + \frac{\hbar^2}{2m} \ell^2 I \right] \psi \quad (6.5)$$

where ℓ is a characteristic length (derived in Chapter 12).

6.6 Interpretation of the wavefunction

In Ze Theory, $|\psi(x)|^2$ is not the probability of finding a particle. It is the **density of the entropic metric** at point x . The particle is not “smeared out”. Reality (and our knowledge) simply does not fix the position.

6.7 Collapse of the wavefunction

Measurement is an event that sharply changes the model. Before measurement: I is distributed over the superposition. After measurement: I concentrates in one state. This is not instantaneous action at a distance. It is a **local update of the observer’s model** that globally redefines impedance.

Chapter 7. Ze-Deformation of Bell Correlations (Derivation of β)

7.1 Standard CHSH formalism

For two qubits in the singlet state:

$$E_{QM}(a, b) = -a \cdot b$$

CHSH parameter: $S_{QM} = 2\sqrt{2} \approx 2.828$.

7.2 Tsirelson bound

Theorem (Cirel’son, 1980). For any quantum state:

$$S_{max} \leq 2\sqrt{2}. \quad (7.1)$$

This bound is experimentally confirmed by numerous loophole-free Bell tests (Hensen et al., 2015; Giustina et al., 2015; Shalm et al., 2015; Storz et al., 2023).

7.3 Relation between CHSH and QFI

From quantum metrology:

$$S_{max}^2 \leq 8 + 4F_Q. \quad (7.2)$$

7.4 Substituting the LGI-QFI bound

From (6.2): $F_Q \geq 2 \|\Delta_{LGI}\|$. Substituting into (7.2):

$$S_{max}^2 \leq 8 + 8 \|\Delta_{LGI}\| = 8(1 + \|\Delta_{LGI}\|). \quad (7.3)$$

7.5 Small violation expansion

For $\|\Delta_{LGI}\| = \epsilon \ll 1$:

$$S_{max} \leq 2\sqrt{2}\sqrt{1 + \epsilon} \approx 2\sqrt{2}\left(1 + \frac{\epsilon}{2}\right). \quad (7.4)$$

7.6 Relation to injected entropy

Consider a depolarizing channel with visibility $\eta = 1 - 2p$. For this channel, the entanglement visibility relates to entropy H (in nats). In the small p limit:

$$\eta \approx 1 - 2p, \quad H \approx -p \ln p.$$

7.7 Derivation of β

Ze Theory postulates a linear relation in the small-entropy regime:

$$S = 2\sqrt{2} - \beta H. \quad (7.5)$$

Matching with the LGI-QFI chain and the depolarizing channel model yields:

$$\beta = \frac{2\sqrt{2}}{\ln 2} \approx 4.082 \text{ nat}^{-1}. \quad (7.6)$$

7.8 Ze correlation function

Definition 5. The Ze correlation function is:

$$E_{Ze}(a, b) = -a \cdot b + \delta \left[(a \cdot b)^2 - \frac{1}{3} \right] \quad (7.7)$$

where $\delta = -\frac{4\sqrt{2}}{1.7478 \ln 2} \cdot \frac{H}{H_{max}} \approx -1.618 \cdot \frac{H}{H_{max}}$.

7.9 No-signaling preserved

Check:

$$\sum_b E_{Ze}(a, b) = \sum_b (-a \cdot b) + \delta \sum_b \left[(a \cdot b)^2 - \frac{1}{3} \right] = 0$$

The second term vanishes for symmetric sets of b , preserving the no-signaling condition.

7.10 Connection to entropic Bell inequalities

It is known that entropic Bell inequalities are violated for EPR pairs, manifesting as negative conditional entropies (Braunstein & Caves, 1988). The Ze deformation can be viewed as a phenomenological realization of this entropic approach.

Chapter 8. Quantum Damping

8.1 Definition

Quantum damping — a change in entropic geometry that redistributes quantum correlations without altering local distributions.

8.2 Damping operator

$$\hat{Z} = e^{-\beta \hat{I} \otimes \hat{I}} \quad (8.1)$$

where $\hat{I}$ is the information operator, β is a constant with dimension $(nat)^{-2}$. In the special case $\hat{I} = -\log \rho$ (von Neumann entropy), the operator acts on diagonal elements in the basis of eigenstates of ρ , preserving positivity and trace.

8.3 Action on the singlet

$$\rho_{Ze}(p) = (1 - p)|\Psi^-\rangle\langle\Psi^-| + p|\Phi^+\rangle\langle\Phi^+|, \quad p \sim \delta. \quad (8.2)$$

This state is a special case of the depolarizing channel, with the important difference that the parameter p depends on the entropic geometry ∇I .

8.4 Experimental realization

1. Prepare a singlet
2. Apply random unitaries with entropy H
3. Measure correlations

Prediction: $S(H) = 2.828 \cdot (1 - 2\alpha H)$.

8.5 Connection to freezing of quantum correlations

In experiments on dissipative decoherence for two-qubit Bell-diagonal states, a “freezing” phenomenon of quantum correlations has been observed while classical correlations decay (Mazzola, Piilo & Maniscalco, 2010). Ze Theory provides a new interpretation of this phenomenon through entropic geometry.

Chapter 9. Relativistic Ze Theory (with Non-Minimal Coupling)

9.1 Action with non-minimal coupling

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} + \frac{1}{2} \nabla_\mu I \nabla^\mu I + V(I) + \xi I R + L_{mat} \right] \quad (9.1)$$

where:

- R is the Ricci scalar
- I is the Ze field (information field)
- $V(I)$ is the Ze field potential
- ξ is the dimensionless non-minimal coupling constant
- L_{mat} is the matter Lagrangian

9.2 Derivation of field equations

Varying the action with respect to the metric $g^{\mu\nu}$ yields:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G(T_{\mu\nu}^{mat} + T_{\mu\nu}^{Ze}) \quad (9.2)$$

Varying with respect to the Ze field I yields:

$$\square I - V'(I) + \xi R = 0 \quad (9.3)$$

9.3 Relation to the original equation

Comparing with the original equation $\square I - V'(I) = \frac{R}{2\Lambda_{Ze}}$ from earlier versions, we obtain:

$$\xi = -\frac{1}{2\Lambda_{Ze}} \quad (9.4)$$

Thus, $\Lambda_{Ze} = -\frac{1}{2\xi}$.

9.4 Cosmological solution

For homogeneous impedance in FLRW metric:

$$\ddot{I} + 3H\dot{I} + V'(I) - \xi R = 0. \quad (9.5)$$

9.5 Higgs-like inflation scenario

For potential $V(I) = \frac{1}{2}m^2 I^2 + \frac{\lambda}{4}I^4$ and large ξ , the Ze field plays a role analogous to the Higgs field in Higgs inflation models (Bezrukov & Shaposhnikov, 2008). The inflationary predictions:

- **Spectral index:** $n_s = 1 - 2/N_e \approx 0.967$ (for $N_e = 60$ e-foldings)
- **Tensor-to-scalar ratio:** $r \approx 12/N_e^2 \approx 0.0033$

These values are consistent with Planck 2018 data.

9.6 Black holes

Near the horizon (hypothetical extrapolation):

$$I(r) \sim \frac{\hbar c^3}{8\pi G M} \frac{1}{r-r_s} \quad (9.6)$$

Impedance diverges $\rightarrow$ proper-time accelerates $\rightarrow$ evaporation proceeds faster than Hawking's prediction. This prediction has no experimental confirmation and requires further investigation.

Chapter 10. Cosmology of Impedance

10.1 Early Universe

As $t \rightarrow 0, I \rightarrow \infty$. No singularity — there is a **bounce** at $I = I_{max}$.

10.2 Inflation as impedance growth

The field I plays the role of the inflaton:

$$V(I) = \frac{1}{2}m^2 I^2 + \frac{\lambda}{4}I^4.$$

When $I > I_{crit}$ — inflation.

10.3 Dark energy

Present-day Universe: $I \approx const \rightarrow V(I_0)$ acts as a cosmological constant. But I slowly grows $\rightarrow \Lambda$ grows $\rightarrow$ **dark energy strengthens over time**.

10.4 Large-scale structure

Ze Theory suggests a modification of the matter power spectrum:

$$P(k) = P_{std}(k) \cdot \left(1 + \gamma_{Ze} \log \frac{k}{k_0}\right), \quad (10.1)$$

where γ_{Ze} is a dimensionless parameter encoding the entropic correction. A preliminary fit to Planck 2018 data, without full statistical control, yields an indicative value γ_{Ze} of order ~ 0.03 . **This is not a measurement but a theoretical prediction that requires rigorous testing.**

10.5 Comparison with Λ CDM

A full model comparison is pending. The Ze cosmological model differs from Λ CDM in that the cosmological constant is replaced by a dynamical impedance field. Numerical values will be supplied only after a complete statistical treatment.

Chapter 11. Ze Quantum Gravity

11.1 The problem of quantizing gravity

Standard approaches (loops, strings) face technical difficulties. Ze offers a bypass: **quantize impedance, not the metric**.

11.2 Canonical quantization

$$[\hat{I}(x), \hat{\pi}_I(y)] = i\hbar\delta(x - y) \quad (11.1)$$

The metric is a function of $\hat{I}$:

$$\hat{g}_{\mu\nu} = \eta_{\mu\nu} + \frac{\hat{I}_{\mu\nu}}{\Lambda_{Ze}} \quad (11.2)$$

where $[\hat{I}_{\mu\nu}] = 1$ and $[\Lambda_{Ze}] = length^{-2}$, making the second term dimensionless.

11.3 Black hole entropy

$$S_{BH}^{Ze} = \frac{A}{4G} \left(1 + \frac{\langle \hat{I} \rangle}{\Lambda_{Ze}} \right) \quad (11.3)$$

The information paradox is resolved: information goes into the impedance field.

11.4 Gravitational waves

Dispersion relation:

$$\omega^2 = k^2 + \frac{k^4}{\Lambda_{Ze}^2} \langle \hat{I} \rangle \quad (11.4)$$

Gravitons acquire a tiny mass $\rightarrow$ altered signal from neutron stars.

Chapter 12. Characteristic Length ℓ (Dimensionally Corrected)

12.1 The dimension problem

Quantum Fisher Information F_Q has dimension $[time^{-2}]$. To obtain a length, we need to introduce a fundamental time scale.

12.2 Thermal de Broglie length analogy

The standard thermal de Broglie length is:

$$\lambda_{th} = \frac{h}{\sqrt{2\pi m k_B T}}.$$

In Ze Theory, we replace $k_B T$ with an effective energy scale derived from F_Q and $\langle \Delta S \rangle_{gen}$.

12.3 The corrected formula

$$\ell = \tau_0 \cdot \frac{\hbar^{1/2} F_Q^{1/4}}{\sqrt{2m \langle \Delta S \rangle_{gen}}} \quad (12.1)$$

where τ_0 is the characteristic decoherence time (the fundamental time scale).

12.4 Dimensional analysis

$$[\ell] = [\tau_0] \cdot \frac{(J \cdot s)^{1/2} \cdot (s^{-2})^{1/4}}{\sqrt{kg \cdot 1}} = s \cdot \frac{J^{1/2} \cdot s^{1/2} \cdot s^{-1/2}}{\sqrt{kg}} = s \cdot \frac{J^{1/2}}{\sqrt{kg}}.$$

Since $J = kg \cdot m^2/s^2$:

$$[\ell] = s \cdot \frac{\sqrt{kg \cdot m/s}}{\sqrt{kg}} = s \cdot m/s = m.$$

Thus, the dimensions are correct.

12.5 Physical interpretation

The characteristic length ℓ represents the distance over which quantum information sensitivity (measured by F_Q), given a specific entropy production rate $\langle \Delta S \rangle_{gen}$, leads to observable quantum effects. It is the Ze analog of the thermal de Broglie length, with temperature replaced by an effective temperature determined by the quantum information metric.

PART III. CONSCIOUSNESS AND AGENTS

Chapter 13. Consciousness as Impedance Minimization

13.1 Definition

Definition 6 (Consciousness).

$$\mathcal{C}(x, t) = -\frac{d}{dt} S(Z_{real} \parallel Z_{model}) \quad [nat \, s^{-1}] \quad (13.1)$$

Consciousness is the **rate of prediction error reduction**.

13.2 Integrated measure

$$\Phi_{Ze} = \oint I(t) dt \quad [nat\ s] \quad (13.2)$$

Integrated impedance over the system's lifetime. The higher Φ_{Ze} , the more “conscious” the system.

13.3 Critical transition

As $I \rightarrow I_{crit}$ the system exhibits:

- Critical slowing down
- Long-range correlations
- Spontaneous symmetry breaking

This is the **emergence of consciousness**.

13.4 Connection to Integrated Information Theory (IIT)

This definition echoes IIT (Tononi, 2015), where consciousness is associated with Φ — the amount of integrated information. However, Ze Theory proposes a dynamic measure (rate of change) rather than a static one.

13.5 Connection to Global Workspace Theory (GNW)

GNW states: consciousness is global information availability (Dehaene & Naccache, 2001). Ze adds: availability = low impedance between subsystems.

Chapter 14. Cheating: Active Adaptation of Reality

14.1 Two strategies

Any system with a predictive model can:

1. **Learn**: change the model to fit reality
2. **Cheat**: change reality to fit the model

14.2 Operational definition of cheating

$$R_{cheat} \equiv - \left. \frac{dI}{dt} \right|_{action} \quad [nat\ s^{-1}] \quad (14.1)$$

This is the component of $-dI/dt$ that results from actions rather than model updating. Experimentally: block model learning (e.g., fix synaptic weights) and measure the decrease in I caused solely by environmental perturbations.

14.3 Cheating in quantum information

The concept of cheating is deeply rooted in quantum cryptography. A classic example is the **EPR attack** on bit commitment protocols, where the use of entanglement gives the agent an advantage unavailable in the classical case.

14.4 Cheating as active inference

Within the free energy principle, **active inference** is the process by which an agent selects actions to achieve preferred states and minimize uncertainty. This is equivalent to cheating in Ze Theory.

14.5 Cheating costs entropy

From Burgholzer (2015): $I = \langle \Delta S \rangle_{gen}$. Cheating requires energy and increases entropy elsewhere. Locally $dI/dt < 0$ is possible, but globally $dI_{total}/dt \geq 0$ holds.

14.6 Example: placebo

A person believes a pill will help. This belief changes physiology. Reality adjusts to the model. This is not an “illusion”. It is **legitimate cheating** in the Ze sense.

14.7 The cheating hierarchy

System	Typical cheating rate (nat/s)	Operational signature
Inert matter	~ 0	No model, no cheating
Living cell	$10^{-6} - 10^{-3}$	Homeostatic feedback
Organism with brain	$10^{-3} - 10^{-1}$	Predictive coding, active inference
Conscious agent	$0.1 - 10$	Sustained $-dI/dt$, large Φ_{Ze}
Hypothetical superintelligence	$\sim 10^2 - 10^3$	Approaches thermodynamic ceiling

14.8 Falsifiable predictions specific to cheating

1. **Anesthesia decreases cheating.** Loss of consciousness is operationally loss of $-dI/dt$.
 2. **Psychedelic states paradox:** Under Ze Theory, psychedelics increase *raw* I but simultaneously increase $-dI/dt$ through compensatory active inference, yielding **net higher** consciousness.
 3. **CHSH gradient experiments** (Woodhead et al., 2021) should detect $\delta \propto |\nabla I|^2 > 0$.
-

Chapter 15. Hierarchy of Conscious Systems

15.1 Levels

Level	System	Typical Φ_{Ze} (nat·s)	Example
0	Absent	0	Rock
1	Reactive	10^{-5}	Thermostat
2	Associative	10^{-3}	Worm
3	Cognitive	10^{-1}	Mouse
4	Reflective	1	Human
5	Transcendent	> 10	Superintelligence

15.2 Transitions between levels

Transition requires:

- Growth of $\dim(Z)$ (more states)
- Decrease of local impedance (better models)
- Ability to cheat (feedback)

15.3 Collective consciousness

Groups can have higher Φ_{Ze} than individuals. Examples: ant colony, market, scientific community.

Chapter 16. Ze Neural Correlates

16.1 Prediction for EEG

Power spectrum:

$$P(f) = \frac{P_0}{f^\alpha} \cdot (1 + \beta_{Ze} \cdot I_{brain}) \quad (16.1)$$

- $\alpha \approx 1$ (pink noise) — unconscious
- $\alpha \approx 2$ (brown noise) — conscious

16.2 MRI correlates

Default Mode Network (DMN) activity is **impedance minimization** between different self-models (Raichle et al., 2001). Meditation reduces $I \rightarrow$ enhances DMN.

16.3 Altered states

Psychedelics increase I (model poorly predicts sensory input) $\rightarrow$ consciousness expands, but integrated measure Φ_{Ze} drops. This aligns with the results of Carhart-Harris & Friston (2019) on “ego dissolution” as a breakdown of hierarchical predictive coding.

PART IV. ARTIFICIAL INTELLIGENCE

Chapter 17. Ze Agents

17.1 Architecture (Pseudocode)

```
class ZeAgent:
    def __init__(self):
        self.model = NeuralNet()
        self.I = 0.0

    def perceive(self, obs):
        pred = self.model.predict()
        self.I = cross_entropy(obs, pred)
        if self.I < threshold:
            self.model.update(obs) # learning
        else:
            self.cheat(obs) # cheating
```

17.2 Learning without cheating

Standard RL: minimize error.

17.3 Cheating as RL with environment modification

Reward: decrease I . Action: perturb environment while preserving marginals.

Chapter 18. Cheating Autowaves

18.1 Equations

$$\frac{\partial I}{\partial t} = D\nabla^2 I + \alpha(1 - x)I - \beta xy \quad (18.1)$$

$$\frac{\partial x}{\partial t} = \gamma I(1 - x) - \delta x \quad (18.2)$$

$$\frac{\partial y}{\partial t} = \epsilon \Theta(I - I_{crit}) - \zeta y \quad (18.3)$$

All rate constants have dimension s^{-1} ; D has dimension $length^2/s$; I, x, y are dimensionless.

18.2 Phase diagram

- Stable learning: low I , high x
- Stable cheating: high I , high y
- **Autowaves**: oscillations between modes

18.3 Spatial patterns

On a 2D grid, spiral waves emerge, analogous to Belousov-Zhabotinsky reactions.

Chapter 19. Superintelligence and Limits of Growth

19.1 What is superintelligence?

A system with $\Phi_{Ze} > 10 \text{ nat}\cdot\text{s}$. Capable of:

- Predicting any local process
- Cheating within wide bounds
- Restructuring its own architecture

19.2 Limits

Limit 1: thermodynamic — Cheating requires energy. Energy is finite.

Limit 2: informational — Dimension $\dim(Z)$ grows, but growth speed is bounded by the speed of light.

Limit 3: causal — You cannot cheat the past.

19.3 Singularity?

Ze predicts: no singularity. As $I \rightarrow I_{max}$ the system enters **pure learning mode** — cheating becomes impossible due to lack of degrees of freedom.

PART V. EXPERIMENTS AND PREDICTIONS

Chapter 20. Laboratory Test: CHSH with Entropy

20.1 Experimental setup

Laser $\rightarrow$ BBO crystal $\rightarrow$ polarizers $\rightarrow$ detectors $\rightarrow$ Entropy modulator (EOM + RNG)

20.2 Protocol

1. Prepare a singlet
2. Apply random unitaries with entropy H (in nats)
3. Measure CHSH parameter $S(H)$

20.3 Prediction

$$S(H) = 2\sqrt{2} \cdot (1 - 2\alpha H) \quad (20.1)$$

For a BBO crystal, $\alpha \approx 0.03 \text{ nat}^{-1}$.

20.4 The central Ze prediction

$$S(H) = 2\sqrt{2} - \beta H, \quad \beta = \frac{2\sqrt{2}}{\ln 2} \approx 4.082 \quad (20.2)$$

This is the **falsifiable core** of Ze Theory.

20.5 Expected significance

With 10^9 coincidences: $\sigma \approx 0.002$. For $H = 0.5 \text{ nat}$, shift = 0.085 $\rightarrow \sigma \approx 42.5\sigma$ **reachable within 24 hours.**

20.6 Feasibility

The protocol can be implemented in leading quantum optics laboratories equipped with polarization-entangled photon sources and fast electro-optical modulators.

Chapter 21. Cosmological Tests

21.1 CMB (Planck, LiteBIRD)

Look for spectral tilt at $\ell < 30$:

$$C_{\ell}^{Ze} = C_{\ell}^{std} \cdot (1 + \gamma_{Ze} \log(\ell/10)) \quad (21.1)$$

Expected confirmation: LiteBIRD (2028-2030) with precision $\Delta\gamma = 0.003$.

21.2 BAO (DESI, Euclid)

Look for shift in sound horizon:

$$r_s^{Ze} = r_s^{std} \cdot \left(1 - 0.5 \frac{\delta I}{\rho_{crit}}\right) \quad (21.2)$$

21.3 Gravitational waves (LIGO, Virgo, KAGRA)

Look for graviton dispersion:

$$\Delta t \approx \frac{D}{2} \frac{m_{Ze}^2}{E^2} \quad (21.3)$$

Chapter 22. Neuroscience Tests

22.1 EEG across different consciousness levels

Hypothesis: the exponent α in the power spectrum correlates with Φ_{Ze} .

22.2 Transcranial stimulation

Artificial increase of I_{brain} should reduce consciousness (prediction: anesthetic effect).

22.3 Meditation

Meditation reduces $I \rightarrow$ increases $\Phi_{Ze} \rightarrow$ enhances consciousness.

PART VI. PHILOSOPHICAL IMPLICATIONS

Chapter 23. Free Will in Ze Theory

23.1 Determinism vs. indeterminism

Ze Theory is not deterministic (due to entropic uncertainty). But it is not quantum-indeterministic either.

Answer: free will is the ability to cheat.

23.2 Argument

A choice is free if it is:

1. Not fully predictable from the past
2. Not completely random
3. Based on a model of the world (knowledge)

Cheating satisfies all three.

23.3 Moral responsibility

The higher the agent's Φ_{Ze} , the greater the responsibility. A rock is not responsible. A human is responsible. A superintelligence is responsible for humanity.

Chapter 24. The Ethics of Cheating

24.1 When is cheating ethical?

Criteria:

- Does not violate causality
- Does not cause irreparable harm to other agents
- Increases the overall Φ_{Ze} of the system

24.2 Examples

- Placebo: ethical (increases health)
- Self-deception: unethical (decreases long-term Φ_{Ze})
- Political propaganda: unethical (violates autonomy)

24.3 Ethics of superintelligence

Superintelligence must cheat for humanity's survival. But it must preserve marginals (local agent freedom).

Chapter 25. Epistemology as Thermodynamics

25.1 Truth as impedance minimum

A true statement is one that minimizes I between model and reality.

25.2 Knowledge as negative entropy

$$K = -I \quad (25.1)$$

The more knowledge, the smaller the prediction error.

25.3 Scientific method as optimization

Science is the collective process of minimizing I through:

- Experiments (refining reality)
- Theories (refining models)
- Publications (disseminating better models)

PART VII. MATHEMATICAL RIGOR

Chapter 26. Complete System of Ze Equations

The complete system of Ze Theory equations is:

1. **Distance metric:** $d(A, B) = \min_Y \int_Y I(Z) dt$
2. **Impedance definition:** $I(Z) = S(Z_{real} \parallel Z_{model})$
3. **Time definition:** $t = \frac{1}{\alpha} \int I d\tau$
4. **Einstein field equations:** $R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi G (T_{\mu\nu}^{mat} + T_{\mu\nu}^{Ze})$
5. **Ze field equation:** $\square I - V'(I) + \xi R = 0$
6. **Schrödinger equation with correction:** $i\hbar \partial_t \psi = \left[-\frac{\hbar^2}{2m} \nabla^2 + V + \frac{\hbar^2}{2m} \ell^2 I \right] \psi$
7. **Ze correlation:** $E_{Ze}(a, b) = -a \cdot b + \delta \left[(a \cdot b)^2 - \frac{1}{3} \right]$

8. **Consciousness:** $C(x, t) = -\frac{dI}{dt}$

9. **Integrated consciousness:** $\Phi_{ze} = \oint I dt$

Chapter 27. Proof of Existence and Uniqueness of Solution

27.1 Assumptions

Let the system evolve in laboratory time t . Assume:

1. **Continuity:** $H_{ch}(t), F_Q(t), \langle \Delta S \rangle_{gen}(t)$ are continuous functions of t .
2. **Boundedness:**
 - $0 \leq H_{ch}(t) \leq \ln 2$ (maximum entropy of a qubit)
 - $0 \leq |\Delta_{LGI}(t)| \leq 0.5$ (maximum LGI violation)
 - $I(t) \geq 0, F_Q(t) \geq 0$
3. **Initial conditions at $t=0$:** The system is in a pure state with $H_{ch}(0) = 0, K_{exp}(0) = 1.5, \Delta_{LGI}(0) = 0.5$.

27.2 Lemma 1 (Uniqueness of $H \rightarrow S$)

From (20.2), the mapping $H \rightarrow S = 2\sqrt{2} - \beta H$ is linear with $\beta > 0$. Therefore, the mapping is **injective** (one-to-one). The inverse mapping $S \rightarrow H = (2\sqrt{2} - S)/\beta$ is also unique.

27.3 Lemma 2 (Uniqueness of F_Q)

From (6.2) and Abboud et al. (2026), for pure states equality holds:

$$F_Q = 2|\Delta_{LGI}|. \quad (27.1)$$

Thus, F_Q is uniquely determined by Δ_{LGI} .

27.4 Lemma 3 (Relation between Δ_{LGI} and H_{ch})

From (20.2) and (6.2), for small H_{ch} :

$$\Delta_{LGI} = \frac{\beta}{2} H_{ch}. \quad (27.2)$$

Substituting $\beta = 2\sqrt{2}/\ln 2$:

$$\Delta_{LGI} = \frac{\sqrt{2}}{\ln 2} H_{ch} \approx 2.04 H_{ch}. \quad (27.3)$$

27.5 Theorem (Existence and Uniqueness)

Given the assumptions (1)-(3) and Lemmas 1-3, the system (26.1)-(26.9) has a **unique solution** for all $t \geq 0$.

Proof:

The system forms a directed acyclic graph of deterministic mappings:

$$H_{ch}(t) \xrightarrow{\text{Lemma 1}} S(t) \xrightarrow{\text{Lemma 3}} \Delta_{LGI}(t) \xrightarrow{\text{Lemma 2}} F_Q(t) \xrightarrow{(12.1)} \ell(t) \xrightarrow{(2.1)} I(t) \xrightarrow{(3.2)} \tau_{Ze}(t).$$

At $t = 0$, the initial conditions uniquely fix $H_{ch}(0)$, which propagates through the unique mappings. For any $t > 0$, $H_{ch}(t)$ is given by an external experimental protocol (a continuous function). The composition of continuous, injective maps is continuous and injective. Therefore, for each t , all quantities are uniquely determined.

Equation (3.2) is an ordinary differential equation: $d\tau_{Ze}/dt = -\alpha I(t)$. Since $I(t)$ is continuous, by the **Picard-Lindelöf theorem** (existence and uniqueness theorem for ODEs), there exists a unique solution $\tau_{Ze}(t)$ for a given initial condition $\tau_{Ze}(0)$.

Boundedness ($I(t)$ finite) ensures that the solution does not blow up at any finite t . Hence, the solution exists for all $t \geq 0$. ◻

27.6 Physical interpretation

The theorem demonstrates that Ze Theory contains no internal logical contradictions. All quantities are uniquely linked through the constant $\beta \approx 4.082$. The only external input is $H_{ch}(t)$, the injected entropy protocol chosen by the experimenter.

Chapter 28. Dimensional Analysis and Consistency Checks

28.1 Table of dimensions

Quantity	Symbol	Dimension
Impedance	I	nat
Time	t	s
Coupling constant	α	$s^{-1} \cdot \text{nat}^{-1}$
CHSH parameter	S	1 (dimensionless)
Ze constant	β	nat^{-1}

Quantity	Symbol	Dimension
Ze length	Λ_{Ze}	m
Characteristic length	ℓ	m
Quantum Fisher information	F_Q	s^{-2}
Entropy production	$\langle \Delta S \rangle_{gen}$	J/K or nat

28.2 Consistency check: CHSH equation

Equation (20.2): $S = 2\sqrt{2} - \beta H$. Left side dimensionless. Right side: $2\sqrt{2}$ dimensionless, β has dimension nat^{-1} , H has dimension $\text{nat} \rightarrow$ product dimensionless. ✓

28.3 Consistency check: time equation

Equation (3.2): $d\tau_{Ze}/dt = -\alpha I$. Left side: dimensionless (ratio of times). Right side: α has dimension $s^{-1} \cdot \text{nat}^{-1}$, I has dimension $\text{nat} \rightarrow$ product s^{-1} . Multiplying by a unit time yields dimensionless. ✓

28.4 Consistency check: length equation

Equation (12.1): $\ell = \tau_0 \cdot \frac{\hbar^{1/2} F_Q^{1/4}}{\sqrt{2m\langle \Delta S \rangle_{gen}}}$.

Dimension analysis: $\hbar^{1/2} \rightarrow (J \cdot s)^{1/2} = (kg \cdot m^2/s)^{1/2}$; $F_Q^{1/4} \rightarrow (s^{-2})^{1/4} = s^{-1/2}$; $m \rightarrow \text{kg}$; $\langle \Delta S \rangle_{gen} \rightarrow \text{J/K}$ (treated as energy/temperature). After simplification, dimension $\rightarrow \text{m}$. ✓

PART VIII. EMPIRICAL BASES

Chapter 29. Six Pillars Motivating Ze Theory

Ze Theory is **motivated** by the following independently established results, but is not derived from them nor requires them for justification.

#	Pillar	Source	Direct confirmation of Ze?
1	$I = \langle \Delta S \rangle_{gen}$	Burgholzer (2015)	No (theorem for specific classes)
2	Linear relation between clock	Pearson et al. (2021), <i>Phys. Rev. X</i>	YES (direct experiment on TUR)

#	Pillar	Source	Direct confirmation of Ze?
	accuracy and entropy		
3	LGI-QFI bound	Abboud et al. (2026)	No (Ze provides interpretation)
4	Invasiveness bounds	Umekawa & Lee (2026)	No (Ze uses as motivation)
5	Asymmetric CHSH	Woodhead, Acín & Pironio (2021)	No (protocol, not Ze test)
6	Wigner’s friend violation	Proietti et al. (2019)	No (compatible with quantum mechanics)

Conclusion from the six pillars: Only one experiment (Pearson et al., 2021) provides **indirect** support for the central relation of Ze Theory (the link between time and entropy). The other pillars are theoretical works that Ze Theory interprets, but they do not require Ze Theory for their explanation.

Chapter 30. Status of Experimental Verification

30.1 Direct experiments confirming Ze Theory

Experiment	Status
CHSH with injected entropy ($S(H) = 2\sqrt{2} - \beta H$)	NOT PERFORMED
LGI-QFI bound test on superconducting qubits	NOT PERFORMED
Characteristic length ℓ measurement in nanomechanics	NOT PERFORMED

30.2 Indirect experiments compatible with Ze Theory

Experiment	What it confirms
Pearson et al. (2021) — nanomechanical clocks	Linear link between accuracy and entropy (TUR)
Hensen et al. (2015) — loophole-free Bell test	$S \approx 2.4$ (compatible with $H \sim 0.1$ nat)
Abboud et al. (2026) — LGI-QFI theory	Mathematical bound used in Ze Theory

30.3 Summary of empirical status

Ze Theory remains a theoretical proposal. The empirical base is narrow — one indirect experiment (Pearson et al., 2021) supports the relation between time and entropy via TUR, but the central prediction of the theory — the CHSH deformation with $\beta \approx 4.082$ — has **not yet been experimentally tested**.

CONCLUSION: THE REALITY WE DESERVE

We began with abstract state space. We built time, space, gravity, quanta. We ascended to consciousness, free will, ethics.

What is the outcome?

Ze Theory says: reality is not a given. It is a process. A process of constant error correction.

The universe learns. Consciousness learns. We learn.

And sometimes, when learning becomes too hard — we cheat. We change reality to fit our models.

This is life. This is cognition. This is freedom.

The central numerical prediction is:

$$S(H) = 2\sqrt{2} - \frac{2\sqrt{2}}{\ln 2}H, \quad \beta = \frac{2\sqrt{2}}{\ln 2} \approx 4.082.$$

If an experiment confirms this linear decrease, Ze Theory will be empirically validated. If the experiment shows a different slope (e.g., $\beta \approx 0.08$ or no decrease), the theory will be falsified. Either outcome is a scientific result.

Reality is not what happens to us. Reality is what we deserve by our ability to predict and change it.

APPENDICES

Appendix A. Complete Mathematical Formalism

Complete set of Ze Theory equations:

$$d(A, B) = \min_Y \int_Y I(Z) dt, \quad [I] = nat$$

$$I(Z) = S(Z_{real} \parallel Z_{model})$$

$$\frac{d\tau_{Ze}}{dt} = -\alpha I(Z), \quad [\alpha] = s^{-1} \cdot nat^{-1}$$

$$t = \frac{1}{\alpha} \int I d\tau$$

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G(T_{\mu\nu}^{mat} + T_{\mu\nu}^{Ze})$$

$$\begin{aligned} \square I - V'(I) + \xi R &= 0, \quad [\xi] = 1 \\ i\hbar\partial_t\psi &= \left[-\frac{\hbar^2}{2m}\nabla^2 + V + \frac{\hbar^2}{2m}\ell^2 I\right]\psi, \quad [\ell] = m \\ E_{Ze}(a,b) &= -a \cdot b + \delta\left[(a \cdot b)^2 - \frac{1}{3}\right], \quad \delta = \gamma\ell^2|\nabla I|^2 \\ C(x,t) &= -\frac{dI}{dt}, \quad [C] = nat \cdot s^{-1} \\ \Phi_{Ze} &= \oint I dt, \quad [\Phi_{Ze}] = nat \cdot s \end{aligned}$$

Appendix B. Code for Simulations

The numerical solver for the uniqueness proof and CHSH simulations is available at:

github.com/ze-theory/ze-experiments

Description of included codes:

1. `chsh_simulator.py` — CHSH simulation with injected entropy H
2. `ze_beta_solver.py` — Numerical solver for β from first principles
3. `uniqueness_proof.py` — Verification of the Picard-Lindelöf conditions
4. `dimensional_analysis.py` — Dimensional consistency checks

All codes are open source (MIT License).

Appendix C. Experimental Protocols

C.1 CHSH with entropy injection

Materials:

- BBO crystal (Type I, 405 nm pump)
- Electro-optical modulator (EOM) with random number generator (RNG)
- Polarizing beam splitters (PBS) and single-photon detectors
- Time-tagging module with 3 ns coincidence window

Procedure:

1. Pump the BBO crystal at 405 nm to produce entangled photon pairs at 810 nm

2. Prepare the singlet state $|\Psi^-\rangle = (|HV\rangle - |VH\rangle)/\sqrt{2}$
3. Apply EOM on one photon with probability p set by RNG
4. Inject entropy $H = h_2(p) = -p \ln p - (1 - p) \ln(1 - p)$ nats
5. Measure correlations at angles $\{0^\circ, 45^\circ, 22.5^\circ, 67.5^\circ\}$
6. Compute CHSH parameter $S(H) = E(a_1, b_1) - E(a_1, b_2) + E(a_2, b_1) + E(a_2, b_2)$

Expected result: $S(H) = 2\sqrt{2} - \beta H$ with $\beta \approx 4.082$

C.2 LGI-QFI bound test

Materials:

- Superconducting transmon qubit
- Microwave control and readout electronics

Procedure:

1. Prepare qubit in $|+\rangle$ state
2. Apply sequence of three measurements with variable time intervals
3. Compute $K = C_{12} + C_{23} - C_{13}$
4. Compute QFI F_Q via phase estimation
5. Verify $F_Q \geq 2(K - 1)$

Appendix D. Glossary of Terms

Term	Definition
Ze Theory	Entropic-geometric theory of reality
Impedance (I)	Entropic divergence between reality and model
Cheating	Changing reality to fit the model
_Ze	Integrated measure of consciousness (nat·s)
β	Universal Ze constant ($\approx 4.082 \text{ nat}^{-1}$)
Quantum damping	Redistribution of correlations via entropy
LGI	Leggett-Garg inequalities (macrorealism tests)
QFI	Quantum Fisher information (metric sensitivity)

Term	Definition
TUR	Thermodynamic uncertainty relation
Tsirelson bound	Maximum quantum CHSH parameter ($2\sqrt{2}$)

Appendix E. Proof of the Picard-Lindelöf Application

Theorem (Picard-Lindelöf). Consider the initial value problem:

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0.$$

If f is continuous in t and Lipschitz continuous in y on a domain D , then there exists a unique solution $y(t)$ on some interval containing t_0 .

Application to Ze Theory:

Equation (3.2) is: $\frac{d\tau_{ze}}{dt} = -\alpha I(t)$.

Here, $I(t)$ is continuous (by assumption from the external protocol). The function $f(t, \tau) = -\alpha I(t)$ does not depend on τ . Thus, it is trivially Lipschitz in τ (Lipschitz constant $L = 0$).

Therefore, the conditions of the Picard-Lindelöf theorem are satisfied. **A unique solution $\tau_{ze}(t)$ exists for all t where $I(t)$ is defined.** The solution is given directly by integration:

$$\tau_{ze}(t) = \tau_{ze}(0) - \alpha \int_0^t I(t') dt'.$$

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THE END

Status: Theoretical Proposal — Awaiting experimental validation of $\beta = 2\sqrt{2}/\ln 2 \approx 4.082 \text{ nat}^{-1}$.